

APEX: AI Platform of Platforms

Cloud-Native AI Agent Architecture | TOGAF 10 + AI-DLC | v4.0 Peer-Reviewed

TOGAF 10 ADM applied end-to-end to design a governed, living AI platform.
Integrates AI-DLC (AI as co-builder) with continuous architecture evolution loops.

v4.0 incorporates full peer-review: reframed narrative, calibrated productivity claims (2-5x not 10-15x), 7 structural fragility mitigations, Decision Explanation Artefact pattern, Agent Gateway, Embedding Compatibility Contract, Data Maturity Gate, Phase Boundary Receipts, DORA change framing, and 15 new requirements.

v4.0 — PEER-REVIEWED EDITION

ORGANISATION

GlobalCorp Enterprises

FRAMEWORK

TOGAF 10 ADM + AI-DLC

DOCUMENT REF

EA-APEX-MASTER-001

CLASSIFICATION

CONFIDENTIAL

PROGRAMME

APEX — AI Platform of Platforms

CLOUD APPROACH

Cloud-Native / Vendor-Agnostic

VERSION

v4.0 | April 2026

STATUS

APPROVED — Peer-Reviewed

SECTION 00

What This Document Really Is

The reframed narrative — bigger than TOGAF alone

Not Just a TOGAF Document

This blueprint combines TOGAF 10 ADM (governance skeleton) with AI-DLC (velocity engine) inside a Platform-of-Platforms model. The result is an enterprise architecture that **co-evolves with AI** rather than describing it from the outside. It is directionally accurate, partially idealistic, and intentionally forward-looking.

Traditional EA	This Document's Approach	The Architectural Shift
Static artefacts updated quarterly	Artefacts machine-generated, continuously refreshed, version-controlled like code	Documents → Living Digital Twin
Sequential phase-gated ADM	Concurrent bolts; tiered streaming governance	Phases → Continuous loops + human gates
Architects author all artefacts	AI co-authors; architects review and ratify	Author → Reviewer and decision authority
AI is a technology component	AI is co-builder of the architecture itself	Subject → Co-participant
Governance is gatekeeping	Governance is streaming with human escalation	Gatekeeping → Streaming

The Honest Thesis

"The future enterprise will run like an AI system — architecture, code, and operations evolving continuously with AI as co-participant. This document is a blueprint for that transition. It is directionally accurate, partially idealistic, and intentionally forward-looking."

Peer-Review Verdict — At a Glance

Assessment Area	Verdict	Confidence
AI-DLC replacing sprints with bolts	REAL — matches AWS, GitHub, Google direction	High
Platform of Platforms for agents	REAL — AWS Bedrock, Vertex AI, OpenAI MCP all heading here	High
Prompts/embeddings as governed data	REAL — critically underappreciated; 100% correct for regulated AI	High
Governance: gatekeeping → streaming	REAL — strongest section of original document	High
Cloud-native vendor-agnostic design	REAL — Kubernetes + OpenTelemetry + OpenAPI is correct long-term bet	High
10-15x productivity claims	OVERSTATED — 2-5x evidence-based; 10x only in narrow cases	High
'Every engineer understands every line'	IDEALISTIC — needs Layered Verification Model to be enforceable	High
23 → 120 agents without sprawl governance	RISKY — agent chaos mirrors microservices sprawl; gateway needed	Medium
Phase B/C/D concurrent without risk isolation	RISKY — compliance auditability requires Phase Boundary Receipts	High
AI elaboration on low-maturity data	DANGEROUS — bad data amplified at AI speed; Data Maturity Gate needed	High
EU AI Act Art.13 with multi-agent black boxes	UNRESOLVED in v3 — DEA pattern added in v4.0	High
DORA 'dynamic capabilities' framing	TENSION — continuous AI evolution needs controlled change framing	High

SECTION 01

AI-DLC: Research Findings & EA Impacts

What is real, what is overstated, calibrated claims for enterprise use

Calibrated Productivity Claims — v4.0 Correction

v3.0 cited '10-15x productivity gains' without qualification. The evidence-based position: 2-5x is consistent and reproducible. 10x occurs only in narrow, well-scoped, greenfield tasks with high-quality semantic context. Business cases must use conservative estimates; actuals tracked quarterly.

Activity	AWS Claim	Evidence-Based Range	Risk if Overstated
Requirements elaboration	10x faster	3-6x in practice	Misses tribal knowledge, political context, and legacy constraints
Architecture design	5x faster	2-4x in practice	AI proposes patterns inconsistent with org standards without tight steering files
IaC / code generation	10-15x faster	3-7x in practice	IaC hallucinations; superficial review at speed; subtle security misconfigurations
Compliance evidence generation	8x faster	3-5x in practice	AI-generated evidence may miss auditor nuance; human sign-off still mandatory
Observability dashboard setup	5x faster	4-5x realistic	Well-scoped task; high-gain claim is credible here
OVERALL programme velocity	10-15x	2-5x sustainable	Over-acceleration increases governance bypass risk and technical debt

The 'Every Engineer Understands Every Line' Principle — Revised

Noble in intent; not maintainable at enterprise scale. At volume, review becomes superficial. v4.0 replaces this with a Layered Verification Model — structural enforcement, not a process commitment.

Layered Verification Model (L1-L5)

Layer	Mechanism	What It Catches	Human Involvement
L1 Static Analysis	Checkov, Semgrep, Trivy, KICS on every commit	Misconfigurations, CVEs, policy violations	Alert-only; engineer reviews flagged items
L2 AI Explanation	AI coding assistant explains every generated block; explanation recorded as artefact	Logic errors, pattern drift from org standards	Engineer confirms AI explanation before approving; explanation retained 12 months
L3 Property Tests	Terratest / pytest-infra; engineer writes property assertions; AI generates test bodies	Behavioural correctness; resource relationship integrity	Engineer owns assertions; AI generates bodies
L4 Canary Deploy	Progressive delivery (Argo Rollouts); SLO-based auto-rollback	Runtime failures not caught statically	SRE monitors; manual override available to SRE Lead only; logged
L5 Drift Detection	GitOps reconciliation (Argo CD / Flux); continuous drift detection	Configuration drift from declared state	Automated alert; architect investigates anomalies

SECTION 02

Structural Fragility Register

Seven fragility points, risks, and v4.0 architectural mitigations

Seven structural fragility points identified in peer review. Each is mapped to the TOGAF phase it most affects and given a concrete v4.0 architectural mitigation.

F-01 — Governance Tempo Mismatch (Phases G/H)

AI-DLC bolts operate in hours. Traditional governance operates in weeks. Under velocity pressure, HITL reviews become bottlenecks — teams bypass them. This is the exact failure mode that caused microservices sprawl. If bolts run faster than governance can absorb, the ARB becomes a rubber-stamp. EU AI Act and DORA obligations get 'met' on paper only.

v4.0 Mitigation: 5-Tier Governance with Inviolable Pipeline Gates

Gate	Trigger	Responsible	SLA	Bypassable?
T1 Auto-Gate	Every commit — IaC scan, RAGAS, DEA completeness, tag compliance	CI pipeline	Synchronous	NO — pipeline blocks merge; no override path
T2 Arch-Gate	Every agent deployment — conformance to approved pattern	Architect on-call	4 business hrs	NO — deployment blocked until approved
T3 ARB-Gate	New SBB, significant deviation, agent topology change	Full ARB	Weekly	Exception: Group CTO written approval + compliance log entry
T4 Reg-Gate	Any EU AI Act High-Risk agent; DORA-reportable change	Compliance Lead	48 hours	NO — no exception pathway exists
T5 Exec-Gate	Budget >threshold; strategic change; programme risk	Steering Committee	Monthly	NO — requires quorum

F-02 — Semantic Context Drift (Phase C / Requirements)

AI-DLC context stored in steering files drifts across sessions, teams, and model versions. AI working from stale context confidently generates non-compliant artefacts with no error signal. At AI speed, the cost of drift compounds faster than at human speed.

v4.0 Mitigation: Context Integrity Protocol

Control	Description	Tooling	Frequency
Context Versioning	Steering files versioned with semver; breaking changes require architect approval	Git + PR review	Every change
Context Diff Review	Before each bolt, AI generates context diff from previous session; engineer reviews before Construction begins	AI session summary	Every bolt start
Requirements Traceability Audit	Every active requirement maps to at least one artefact; orphaned requirements flagged automatically	CI job + ARS tooling	Daily
Session Summary Archive	AI-generated session summary stored as dated artefact; queryable for audit	Object storage + data catalog	Every session end
Forward-Trace Validation	Before any Phase D deployment, AI generates forward-trace from BR to deployed IaC; architect reviews gaps	AI trace tool + ARB	Per deployment

F-03 — Embedding and Model Version Mismatch (Phase C / Data)

When the embedding model is upgraded, the existing vector index becomes semantically misaligned. Retrieval quality degrades silently — agents appear operational but are reasoning from poor context. In regulated decisions this is dangerous: confident outputs grounded in misaligned retrieval.

v4.0 Mitigation: Embedding Compatibility Contract

Control	Description	Implementation
Embedding Model Registry	Each KB records exact embedding model ID and version at index time; stored in data catalog metadata	Data catalog + KB metadata schema
Re-Index Gate	Any embedding model upgrade triggers automatic full re-indexing of all dependent KBs before activation	CI gate; blocks upgrade without re-index confirmation
RAGAS Regression Gate	Before any embedding change, RAGAS scores are baselined; upgrade blocked if faithfulness or context recall regresses by >5%	RAGAS eval suite in CI
Retrieval Quality Dashboard	Continuous monitoring of RAG quality (chunk relevance, faithfulness per agent per day); anomaly detection	OTEL custom metrics + Grafana
90-Day Rollback Window	Previous embedding index retained 90 days post-upgrade to enable rollback without re-indexing from source	Object storage versioning

F-04 — Agent Sprawl (Phase E / Phase H)

Target: 120 agents. The microservices era showed exactly this failure: ease of creation led to undocumented dependencies, debugging nightmares, and operational burden outpacing gains. Agent systems amplify this risk: agents call other agents, creating emergent dependency graphs no human authored and no architecture diagram describes.

v4.0 Mitigation: Agent Lifecycle Governance + Agent Gateway

Control	Description	Implementation
Agent Registry (mandatory)	Every agent registered before production invocation; unregistered agents receive 403 at gateway	Agent Gateway + K8s admission controller
Agent Dependency Map	Every agent declaration lists tool dependencies and sub-agents invoked; visualised as directed graph	Registry schema extension; Grafana graph panel
Agent Health Score	Monthly composite: usage, error rate, cost/decision, human escalation rate; low-health agents flagged	Prometheus metrics + scoring job
Deprecation Register	Agents with <10 invocations/month for 3 consecutive months auto-flagged for decommission review	Scheduled function + ticket system
Emergent Behaviour Alerts	Agent chains exceeding 3 hops OR total latency >10s trigger architect-on-call alert	Distributed tracing span analysis; OTEL
Budget Enforcement	Monthly token/compute budget per agent; 80% = alert; 100% = throttle; 110% = auto-suspend	Cost tag enforcement; budget automation

SECTION 02 cont.

Structural Fragility Register (continued)
Fragility points F-05 through F-07

F-05 — Phase Blurring vs. Regulatory Risk Isolation (Phases B/C/D)

Running Phases B, C, and D concurrently means a wrong data architecture decision propagates into deployed infrastructure before the business architecture is formally approved. TOGAF phase boundaries exist for risk isolation in regulated environments. 'We ran B/C/D concurrently' is difficult to defend when a regulator asks why a deployed agent has no approved Business Architecture artefact.

v4.0 Mitigation: Phase Boundary Receipts

Boundary	Concurrent Rule	Non-Negotiable Exit Criterion	Regulatory Basis
B → C	Phase C data design may begin once BCM draft exists and domain team has reviewed	Phase C data flows touching personal data require Phase B-approved BA artefact before production deployment	GDPR Art.25 (privacy by design requires business intent before data design)
C → D	Phase D IaC scaffolding may begin once application component model is drafted	No Phase D production deployment without Phase C data classification DPO sign-off for any agent touching personal data	GDPR Art.25; EU AI Act Art.9 (risk management system must precede deployment)
D → F	Phase F migration planning may run parallel to Phase D build	No production traffic shift until Phase D security architecture review complete and all T1 automated checks pass	DORA Art.11 (change management controls)

F-06 — Data Maturity as Hidden Prerequisite (Phase B)

AI-DLC auto-generates capability maps from existing documentation. Most enterprise documentation is stale, inconsistent, and incomplete — tribal knowledge is not written down. AI working from bad documents at AI speed produces bad artefacts at AI speed, seeding every downstream phase with structurally flawed assumptions.

v4.0 Mitigation: Data Maturity Gate — Mandatory Before Phase B AI Execution

Check	Assessment	Minimum to Proceed	Action if Below Standard
Documentation currency	AI-assisted date analysis; flag anything older than 18 months	70% of sources dated within 18 months	Human architect runs targeted discovery sessions before AI elaboration begins
Process coverage	AI identifies process steps with no documentation; flags gaps	>80% of value stream stages have at least one documented source	Workshop with domain SMEs to capture tribal knowledge before AI processing
Data lineage	Automated scan of data catalog for entities with no lineage	All personal/sensitive data entities have documented lineage	Data steward sprint; blocked from Phase C until complete
Terminology consistency	AI generates terminology conflict report across sources	< 5 unresolved terminology conflicts in scope	Glossary alignment session with domain leads; conflicts resolved before context build
Architecture debt baseline	Debt register reviewed; critical debt documented	All critical debt items have owner and target date	Critical debt items affecting Phase B scope need accepted mitigations before elaboration

F-07 — Regulatory Blindspots in Multi-Agent Architecture

EU AI Act Art.13 — Transparency in Multi-Agent Black Boxes

The Supervisor-Worker orchestration pattern creates a transparency problem. A regulator asking 'why did agent X make decision Y?' needs a traceable chain: supervisor routing decision, sub-agent RAG retrieval (which chunks, what scores), model invocation (which version, which prompt version), guardrail evaluation, and human review outcome. This chain spans multiple log stores and is not assembled into a human-readable explanation. EU AI Act Art.13 requires 'appropriate' transparency — log archaeology is not appropriate.

v4.0 Mitigation: Decision Explanation Artefact (DEA) Pattern

DEA Component	Content	Source	Format
Request Context	Original request, classification, risk tier, timestamp	Agent gateway access log	JSON structured
Routing Rationale	Supervisor routing decision, selected sub-agent(s), confidence score	Supervisor OTEL trace	Structured text
Knowledge Retrieved	Top-k chunks, similarity scores, KB version, source document refs	RAG retrieval log	JSON array
Model Invocation	Model ID, model version, prompt version hash, token counts, temperature	Model invocation log (immutable)	JSON structured
Guardrail Evaluation	Guardrail policy version, checks run, results, anonymisations applied	Guardrail evaluation log	JSON structured
Output and Decision	Final output, decision classification, confidence, escalation flag	Agent output log	Structured text
Human Review	If escalated: reviewer ID, outcome, rationale, timestamp	HITL portal audit log	Structured text
DEA Assembly	All above assembled by DEA service on-demand or at decision time; completeness validated	DEA generation service	PDF + JSON; 7-year retention

DORA Tension — Dynamic Capabilities vs. Stable ICT Systems

DORA Art.11 requires ICT systems to be stable, predictable, and subject to controlled change. AI agents that 'learn and adapt in motion' are a direct tension with this. Regulators may characterise continuous AI evolution as 'uncontrolled change' unless a formal framing strategy is in place.

DORA Concern	v4.0 Architectural Response
Continuous model updates = uncontrolled change	Every foundation model version change is a formal change event: ARB approval, RAGAS test, canary deploy, compliance record. Frequency does not equal uncontrolled.
Agent behaviour evolution (RLHF/fine-tuning)	Behaviour changes from feedback learning are classified as 'model updates' and follow the same formal change pipeline. No in-production learning without ARB approval.
Knowledge base refresh = unvalidated update	KB refreshes are content change events (not model changes). RAGAS quality gate runs on every refresh. Human sign-off required for regulatory content changes.
Agent orchestration topology changes	Any new agent-to-agent connection is a dependency change requiring ARB review and registry update. Undeclared connections blocked at gateway level.

PRELIMINARY

Preliminary Phase — Foundation

Revised principles incorporating all fragility mitigations

The Preliminary Phase in v4.0 has been updated to incorporate all seven fragility mitigations, calibrated productivity assumptions, and the five operating model transformations required by AI-DLC.

Architecture Principles — APD-001 v3.0

DOC REF: APD-001 | v3.0 | APPROVED

Business Principles

ID	Principle	Statement v4.0	Key Implication
BP-01	AI as Business Capability	Every APEX agent traces to a measurable outcome. AI-DLC gains reported as 2-5x baseline; 10x requires independent evidence before use in business cases	Business cases use conservative estimates; actuals tracked quarterly and reviewed by steering committee
BP-02	Human-in-the-Loop — Non-Negotiable	All agents on sensitive data or high-stakes decisions have an inviolable human escalation path that cannot be bypassed for velocity reasons	HITL gates are pipeline-enforced; any bypass requires written CTO exception logged in compliance register
BP-03	Platform Thinking with Sprawl Controls	No BU procures standalone AI tools if APEX meets the need within 90 days; equally, no agent created without registry entry and health score	ARB approval for procurement AND for new agent creation; agent registry is the source of truth
BP-04 NEW	Data Maturity Before AI Velocity	AI-DLC Phase B cannot begin until the Data Maturity Gate (F-06) is passed for the domain in scope	Data maturity assessment is the first activity of every Pioneer Domain onboarding; it is a programme gate not a guideline

Technology Principles

ID	Principle	Statement v4.0	Key Implication
TP-01	Cloud-Native Open Standards	CNCF-hosted and open-standard components; proprietary cloud SDKs used only where equivalent OSS is not production-ready	All components containerised; OpenAPI for agent interfaces; OTEL for observability; no cloud SDK in core business logic
TP-02	Security by Design	Security architecture sign-off is a Phase D entry gate — no IaC written before CISO sign-off on security design	CISO sign-off before first IaC line per Pioneer Domain; security is not a Phase G afterthought
TP-03	Observability as First-Class	Every agent emits OTEL traces, Prometheus metrics, and structured logs from the first commit; no agent reaches UAT without a Grafana dashboard	Dashboard-as-code generated in bolt 1; observability is part of Definition of Done
TP-04	Cost Transparency with Hard Throttle	Each agent has a tagged cost centre; budget enforced with hard throttle at 110% — not just an alert	Budget throttle is in the agent gateway, not an alert email; cannot be self-overridden by the agent
TP-05	Layered Verification, Not Hero Review	AI-generated IaC passes L1-L5 Layered Verification Model before merge; no single-person review is sufficient at enterprise scale	CI/CD enforces L1-L4 automatically; L2 AI-explanation artefact retained 12 months
TP-06 NEW	Decision Explanation Artefact by Default	Every High-Risk AI agent (EU AI Act Art.6) produces a DEA for every regulated decision	DEA generation service deployed with Pioneer 1; template applied to all subsequent Pioneers

PHASE A

Architecture Vision

Scope, mandate, stakeholders — with AI political blindspot protocol

Architecture Vision Statement — v4.0

"By Q4 2026, GlobalCorp will operate a unified AI Agent Platform (APEX) on cloud-native open standards. Planning baseline: time-to-market reduced from 9.2 months to 5-6 weeks. Aspirational ceiling: 3-4 weeks with AI-DLC once pipeline is proven. Business cases use 9.5 weeks conservative. Full EU AI Act and DORA compliance achieved. Positive NPV over five years. All productivity claims reviewed against actuals quarterly."

The <1 month TTM figure from v3.0 is retained as an aspirational ceiling for pre-approved template agents in H3. It is NOT used in business cases or executive reporting until achieved and evidenced in at least two Pioneer Domains.

Stakeholder Map — AI Political Blindspot Protocol

AI-DLC Mob Elaboration generates stakeholder analyses from written sources. It structurally cannot capture political context. The following protocol must precede every Mob Elaboration session:

Blindspot	What AI Misses	Human Protocol
Power dynamics	Informal power (who actually decides vs. who is nominally responsible)	1:1 interviews with each key player before Mob Elaboration; architect maps informal influence network
Political constraints	Initiatives that cannot succeed for political reasons even if technically sound	Pre-Mob 'political landscape' session with executive sponsor only
Vendor relationships	Existing commercial relationships, exclusivity clauses, relationship equity	Procurement lead reviews all build/buy SBBs before ARB approval
Unwritten rules	Organisational norms, taboo topics, past failed initiatives that create implicit constraints	'Lessons learned' session with long-tenure staff before Phase B begins
Regulatory interpretation	How this specific regulator has interpreted ambiguous rules historically	Regulatory counsel reviews all compliance artefacts before submission

Architecture Constraints — v4.0

Constraint	Source	v4.0 Architecture Impact
Data must remain within contractual jurisdiction	GDPR / PDPA / LGPD	Multi-region; NetworkPolicy blocks cross-border agent calls for personal data at the network layer
All AI models require Model Risk validation	Internal Model Risk Policy	DEA required for High-Risk agents; Model Risk gate in CI/CD is a T4 Reg-Gate
Cloud-native open standards; no proprietary SDK in business logic	TP-01	LLM calls via provider-agnostic adapter; agent logic portable across providers
Zero-downtime deployment for critical services	Enterprise SLA	Argo Rollouts canary; auto-rollback on SLO breach
DORA continuous AI evolution = controlled change	DORA Art.11	Every model update, KB refresh, and topology change is a formal change event with ARB record
EU AI Act High-Risk classification + DEA	EU Reg. 2024/1689	Risk classifier at intake (WP-012); DEA deployed with Pioneer 1

PHASE B

Business Architecture

Capability model with Adaptive Capability governance and calibrated TTM

Phase B v4.0 introduces Adaptive Capabilities — AI-driven capabilities that evolve in motion — and provides distinct governance for them. The value stream is recalibrated with realistic planning baselines.

Business Capability Map v4.0

L1 Capability	Type	APEX	AI-DLC Adaptive?	v4.0 Governance Model
1. AI Strategy & Governance	STATIC	5/5	Partial	Standard ARB quarterly review
2. Agent Development	ADAPTIVE	5/5	YES — AI-DLC Const.	Bolt-level ARB; L1-L5 verification gate
3. Agent Orchestration	ADAPTIVE	4/4	YES	Agent dependency map; topology change = formal change event (DORA)
4. Knowledge Management	ADAPTIVE	4/4	YES	KB refresh = content change event; RAGAS gate; DPO sign-off for sensitive content
5. AI Operations (AIOps)	ADAPTIVE	5/5	YES	AI-DLC Ops; DORA change framing; real-time monitoring
6. Data Integration	STATIC	3/4	PARTIAL	Data Mesh domain governance; quarterly data product review
7. Platform Enablement	STATIC	4/4	YES — portal	Standard deployment cadence; portal changes are standard releases

Value Stream — Calibrated v4.0 (with Planning Baseline)

Planning baseline: 5-6 weeks total. Business case and budget baseline: 9.5 weeks. Aspirational ceiling (H3, proven pipeline): 3-4 weeks. Do not use ceiling in business cases until evidenced across two Pioneer Domains.

Stage	Baseline	APEX Plan	AI-DLC Ceiling	v4.0 Planning Baseline	Key Constraint v4.0
Intake & Qualification	2-4 wks	5 days	2-4 hrs	2 days	F-06 Data Maturity Gate may add 1-2 weeks in early cohorts
Architecture & Design	4-8 wks	10 days	1-3 days	4-5 days	T3 ARB-Gate cannot be bypassed; political blindspot protocol adds time in complex domains
Build & Integrate	8-16 wks	3 wks	1-2 wks	2 wks	L1-L5 verification is non-negotiable; IaC hallucination review adds time; DEA service adds 1 bolt
Validate & Govern	6-10 wks	10 days	3-5 days	7-8 days	T4 Reg-Gate (Model Risk) is inviolable; DEA generation is additional step for High-Risk agents
Deploy & Monitor	2-4 wks	1 wk	1-2 days	3 days	DORA change framing (F-07) adds a compliance record step to every deployment
TOTAL	~9 months	~9.5 wks	~3-4 wks	~5-6 wks	Use planning baseline for budgets; ceiling is programme ambition for H3

PHASE C

Information Systems Architecture

Data + Application — DEA service, Embedding Contract, Agent Gateway

AI-Native Data Entity Governance — v4.0

DOC REF: DAD-APEX-001 | v2.0

Entity	Class	Governance Owner	Version Control	Compatibility Gate	Retention
Prompt Catalog	C-AI	Platform AI Eng + DPO	Git semver; PR mandatory	Prompt change triggers full regression suite	Per source data class
Embedding Store	C-AI	Data Mesh Domain	Embedding model ID + version in catalog metadata	F-03: RAGAS gate on model upgrade; 90-day rollback	90 days post-supersession
RAG Retrieval Log	C2→C4 inherit	Compliance + Audit	Immutable; object lock	None — append-only	7 years (DORA)
Feedback Log	C2→C4 inherit	Model Risk	Immutable; object lock	None — append-only	7 years (DORA)
Agent Memory	Inherits source	Platform Eng + DPO	TTL policy in KV store; 30 min max	PII check on write; TTL enforced at gateway	Session TTL only; no persistence
Model Lineage Record	C3	Model Risk Mgmt	ML model registry (MLflow equiv.)	New version = formal model update event	Model lifecycle + 7 years
DEA (NEW)	C3	Compliance + Legal	Generated at decision; immutable	DEA service validates completeness before store	7 years (EU AI Act + DORA)

Application Architecture v4.0 — New Components: DEA Service + Agent Gateway

Component	Technology	Pattern	SLA	v4.0 Change
APEX Developer Portal	React SPA + API Gateway (Kong/Nginx/cloud)	BFF	99.9% / p99 <500ms	Agent health score visible; DEA query link per agent
APEX Control Plane	Python FastAPI; Kubernetes; 6 microservices	Microservices	99.9% / p99 <500ms	Agent Lifecycle service + Budget Enforcer service added
Agent Orchestration	LangGraph / AutoGen / managed equiv.	Supervisor-Worker	99.95% / p99 <3s	All routing decisions logged to OTEL trace for DEA assembly
Serverless Tool Functions	Knative / OpenFaaS / cloud FaaS	Action Group / Tool Sidecar	99.95% / p99 <1s	Tool invocation logs written to structured log for DEA
DEA Generation Service NEW	Python event-driven; Kubernetes; triggered on regulated decision	Event-driven async	99.9% / <5s	Assembles DEA from all log sources; validates completeness; immutable store
Agent Gateway NEW	Envoy proxy + custom plugin OR APISIX + plugin	Zero-trust entry point	99.99% / p99 <50ms	Registry lookup; budget throttle; unregistered agent block; dep-map logging
Human Review Portal	React + Temporal / Camunda workflow	HITL review workflow	99.5% / SLA by risk tier	HITL outcome written to DEA; DEA marked complete only after HITL if required
LGTM Observability	OpenTelemetry + Prometheus + Grafana + Loki + Tempo	Centralised OTEL collector	99.5%	Agent health score metrics added; emergent behaviour alert rules added (F-04)

PHASE D

Technology Architecture

Cloud-native infra, security, CI/CD v4.0 with Layered Verification

CI/CD Pipeline — v4.0 with L1-L5 Layered Verification

DOC REF: TAD-APEX-001 | v3.0

Stage	Tools	AI-DLC Role	Gate (Bypassable?)
L1 Static Analysis	Checkov, Semgrep, Trivy, KICS, yamllint	AI explains failures and proposes fixes; cannot self-approve	NEVER — pipeline blocks merge; no override
L2 AI Explanation	AI coding assistant explains every generated block; explanation artefact recorded	Explanation retained as PR comment + artefact for 12 months	NEVER — engineer confirms before approving
L3 Property Tests	Terratest, pytest-infra, contract tests	AI generates test bodies; engineer writes property assertions	NEVER — 100% pass required
RAGAS Gate	RAGAS eval suite; custom eval harness	AI generates adversarial test cases; regression vs. baseline	NEVER — >90% score required
Guardrail Tests	NeMo Guardrails test harness; adversarial prompt library	AI generates edge-case prompts to probe guardrail coverage	NEVER — 100% pass required
T4 Model Risk Gate	Ticket-based approval (Jira/Linear/equiv.)	AI generates Model Risk evidence pack draft; human signs	NEVER — human sign-off mandatory; T4 Reg-Gate
L4 Canary Deploy	Argo Rollouts / Flagger; Prometheus SLO monitoring	AI monitors SLOs; triggers auto-rollback on breach	Override: SRE Lead only; every override logged
DEA Validation	DEA generation service completeness check	DEA assembled and validated before agent marked production-ready	NEVER for High-Risk AI agents
Post-Deploy DORA Record	Argo CD drift detection; cost tag validator; DORA change record writer	AI generates stakeholder summary; DORA change record auto-created	NEVER — records are mandatory

PHASES E-H

Opportunities, Migration, Governance & Change — v4.0

Roadmap, SBBs, 5-tier governance, change intelligence

Phase E — Architecture Roadmap v4.0 (ROAD-APEX-001 v3.0)

DOC REF: ROAD-APEX-001 | v3.0

Pioneer 1 failure contingency: if Pioneer 1 delivers >2 weeks late, subsequent Pioneers shift; programme risk reviewed at Pioneer 3 gate.

Work Package	WP	Bolts	Wks	Horizon	Risk Level v4.0
Multi-region K8s, zero-trust IAM, Vault	WP-001	2	4	H1	Low — well-understood pattern
AI Agent Orchestration (framework eval spike, guardrails, KB-001)	WP-002	2	4	H1	Medium — framework lock-in risk; abstraction layer required
CI/CD L1-L5 verification + DEA generation service deploy	WP-003	2	3	H1	Medium — DEA service is new pattern; allow spike time
Security baseline (zero-trust, mTLS, guardrails, Agent Gateway)	WP-004	2	4	H1	Low — automated; well-understood
LGTM observability + agent health score metrics	WP-005	1	2	H1	Low
Pioneer 1 — Risk Scoring Agent (DEA, Model Risk gate, first end-to-end)	WP-006	3	8	H1	HIGH — all gates run for first time; allow buffer
Pioneer 2 — Customer Verification Agent	WP-007	3	7	H1	Medium — builds on Pioneer 1 learnings
Pioneer 3 — Advisory Rebalancing Agent [Programme Risk Gate]	WP-008	2	6	H2	Medium — MiFID II HITL adds complexity
Pioneer 4 — IT Incident Response Agent	WP-009	2	5	H2	Low — fewer regulatory constraints
Pioneer 5 — Model Monitoring & Compliance Agent	WP-010	3	6	H2	Medium — DORA reporting integration
Self-service portal, marketplace, agent health dashboard	WP-011	3	8	H2	Medium — developer experience is subjective
EU AI Act compliance pack, DEA template library, DORA evidence automation	WP-012	2	5	H2	High — regulatory interpretation risk

Phase G — 5-Tier Governance Model v4.0 (CAF-APEX-001 v2.0)

DOC REF: CAF-APEX-001 | v2.0 – F-01 Tempo Mitigation Applied

Tier	Scope	Method	SLA	Bypassable?
T1 Auto	IaC L1 scan; RAGAS; DEA completeness; tag compliance; budget alert	CI pipeline + K8s admission + scheduled jobs	Synchronous	NO — pipeline blocks; no human override path
T2 Arch	Pattern conformance; new data flow; classification audit	Automated flag → architect on-call (Slack/Teams) + 4hr SLA	4 business hrs	NO — deployment blocked; only ARB can release
T3 ARB	New SBB; significant deviation; agent topology change; debt threshold	Weekly ARB; async for urgent items	Weekly	Exception: CTO written approval + compliance log; no silent bypass
T4 Reg-Gate	High-Risk AI (EU AI Act); DORA-reportable change	Compliance lead; no delegation	48 hrs	NO — no exception pathway exists; hardcoded in pipeline

Tier	Scope	Method	SLA	Bypassable?
T5 Exec	Budget >threshold; strategic change; programme risk escalation	Monthly steering; emergency within 48 hrs for DORA	Monthly	NO — requires quorum

Phase H — Change Intelligence v4.0 (ACHG-APEX-001 v2.0)

DOC REF: ACHG-APEX-001 | v2.0 – DORA Continuous Change Framing Applied

Type	AI Monitoring	Human Decision	DORA Framing	Example
T1 Tech	Model release monitor; OSS feeds; benchmark auto-run vs. GlobalCorp eval	Architect reviews impact; ARB approves	Formal change event regardless of frequency	New open-weight model outperforms by 18%; ARB approves in 3 days with RAGAS evidence
T2 Reg	Regulatory feed parsed daily; impact classified vs. APEX controls	Compliance lead; regulatory counsel if ambiguous	Regulatory change triggers controlled response	EU AI Act Art.6 enforcement: APEX intake auto-classification activated; DORA pack updated
T3 Business	Onboarding request in portal triggers intake workflow	ARB decision within 5 business days	Standard change; ARB record created	New division requests onboarding for contract analysis agent
T4 Simplify	Usage analytics flags agents <10 invocations/month for 3 months	Domain owner: decommission or justify	Decommission = formal change event; IaC teardown + compliance record	Pioneer 1 v1.2 superseded; decommissioned after parallel-run; DEA archive confirmed

APPENDIX A

Honest Assessment Summary Card

What is real, overstated, and resolved — the complete verdict

What This Blueprint Gets Right — Top 5% Thinking in Enterprise AI Today

- ✓ AI-DLC will reshape the entire SDLC — this is happening across AWS, GitHub, and Google internal tooling
- ✓ Architecture will become continuous and living — this is the correct long-term direction for enterprise EA
- ✓ Agents require platform thinking with shared governance, observability, and tooling abstraction — industry is converging here
- ✓ Prompts, embeddings, and feedback logs are first-class governed data — critically underappreciated in most organisations
- ✓ Governance must move from gatekeeping to streaming with human escalation gates — strongest section of the original document
- ✓ Cloud-native vendor-agnostic design (Kubernetes, OpenTelemetry, OpenAPI) is the correct long-term bet for regulated enterprises

What Was Overstated in v3.0 — Corrected in v4.0

- ✗ 10-15x productivity gains → Corrected to 2-5x sustainable; 10x only in narrow well-scoped cases with exceptional data quality
- ✗ 'Every engineer understands every line' → Replaced with L1-L5 Layered Verification Model with structural pipeline enforcement
- ✗ <1 month TTM as planning baseline → Moved to aspirational ceiling; 5-6 weeks is v4.0 planning baseline; 9.5 weeks for business cases
- ✗ Phase B/C/D concurrent without risk isolation → Phase Boundary Receipts added; regulatory data flows require Phase B sign-off before Phase C production deployment
- ✗ Data maturity risk acknowledged but unmitigated → Data Maturity Gate (F-06) is now a mandatory programme gate before Phase B AI execution

Seven Structural Fragilities Resolved in v4.0

Fragility	v3.0 Gap	v4.0 Resolution	New Architecture Component
F-01 Governance Tempo	Governance bypass under velocity pressure	5-tier governance; T1-T5 pipeline-enforced; no silent bypass	CAF-APEX-001 v2.0; CI pipeline gates
F-02 Context Drift	Semantic context drifts across AI sessions	Context Integrity Protocol; session summary archive; forward-trace validation	Context Diff Review process; ARS updates
F-03 Embedding Compat.	Silent RAG quality degradation on model upgrade	Embedding Compatibility Contract; RAGAS regression gate; 90-day rollback	DAD-APEX-001 v2.0; DR-NEW-01/02
F-04 Agent Sprawl	120 agents with no sprawl governance	Agent Gateway; Agent Lifecycle service; health score; budget hard-throttle	AAD-APEX-001 v2.0; Agent Gateway (WP-004)
F-05 Phase Blurring	Concurrent B/C/D without risk isolation	Phase Boundary Receipts; DPO sign-off gate for personal data flows	Updated ADM process; T4 Reg-Gate
F-06 Data Maturity	AI elaboration on stale enterprise data	Data Maturity Gate; 5-check assessment before Phase B	DM-NEW-01; F-06 mitigation table; BP-04
F-07 Regulatory Blindspots	Multi-agent black box; DORA tension	DEA pattern; DORA change framing; all AI changes = formal change events	DEA service; CR-NEW-01/02; ACHG v2.0

APPENDIX B

Cloud-Native Service Mapping

OSS reference implementations and cloud-provider equivalents

All APEX capabilities specified using OSS/open-standard references. Cloud-provider equivalents listed for reference only. Provider selection in Phase F.

APEX Capability	OSS / Open-Standard	Provider A (AWS)	Provider B (GCP)	Provider C (Azure)
Container Orchestration	Kubernetes (CNCF)	EKS	GKE	AKS
Serverless Functions	Knative / OpenFaaS	AWS Lambda	Cloud Functions	Azure Functions
Agent Orchestration	LangGraph / AutoGen / CrewAI	Bedrock Agents	Vertex AI Agents	Azure AI Foundry
Foundation Model API	Ollama / LiteLLM adapter	Bedrock (Claude, Titan)	Vertex AI (Gemini)	Azure OpenAI (GPT-4o)
Vector Store	Weaviate / Qdrant / Milvus	OpenSearch Serverless	AlloyDB + pgvector	Azure AI Search
Object Storage	MinIO (S3 API)	S3	Cloud Storage	Azure Blob Storage
Relational DB	PostgreSQL (K8s managed)	Aurora PostgreSQL	Cloud SQL PG	Azure DB for PostgreSQL
Event Streaming	Apache Kafka / Redpanda	Kinesis / MSK	Pub/Sub	Azure Event Hubs
Workflow Engine	Temporal / Argo Workflows	Step Functions	Cloud Workflows	Azure Durable Functions
Secrets Mgmt	HashiCorp Vault	Secrets Manager + KMS	Secret Manager + KMS	Azure Key Vault
Metrics	Prometheus	CloudWatch Metrics	Cloud Monitoring	Azure Monitor
Tracing	Jaeger / Grafana Tempo (OTEL)	AWS X-Ray	Cloud Trace	Azure App Insights
Logging	Loki / OpenSearch	CloudWatch Logs	Cloud Logging	Azure Monitor Logs
Dashboards	Grafana	CloudWatch Dashboards	Cloud Monitoring	Azure Dashboards
GitOps / CD	Argo CD / Flux	CodePipeline	Cloud Deploy	Azure DevOps
Service Mesh	Istio / Linkerd (CNCF)	App Mesh	Traffic Director	Azure Istio add-on
ML Tracking	MLflow	SageMaker Experiments	Vertex AI Experiments	Azure ML Experiments
AI Safety Layer	NeMo Guardrails / Guardrails AI	Bedrock Guardrails	Vertex AI Safety	Azure AI Content Safety
IaC	Terraform / OpenTofu + Terratest	CDK / CloudFormation	Deployment Manager	Bicep / ARM
Progressive Delivery	Argo Rollouts / Flagger	CodeDeploy	Cloud Deploy canary	Azure Deployment Strategies
Agent Gateway	Envoy + plugin / APISIX	API Gateway + Lambda auth	Apigee + policy	API Management + policy
IaC Security (L1)	Checkov + Semgrep + Trivy	CodeGuru + Inspector	Artifact Registry scan	Defender for DevOps
Runtime Security	Falco (CNCF)	GuardDuty	Security Command Center	Defender for Containers

Document Control — Version History

Version	Date	Author	Change Summary	Approver
1.0	2025-01-15	Enterprise Architect	Initial draft — Phases A-D	ARB
2.0	2025-03-01	Enterprise Architect	Full ADM coverage; Architecture Contracts; Governance sections	Group CTO
3.0	2026-04-01	Enterprise Architect	Cloud-native edition: AWS removed; AI-DLC integrated; table wrapping fixed	Group CTO
4.0	2026-04-15	Enterprise Architect	Full peer-review: reframed narrative; 2-5x calibrated claims; Layered Verification L1-L5; 7 fragility mitigations (DEA service, Agent Gateway, Embedding Contract, Data Maturity Gate, Phase Boundary Receipts, DORA framing, Agent Sprawl controls); 15 new requirements; Honest Assessment appendix; political blindspot protocol; TTM recalibrated with planning baseline vs. aspirational ceiling	Group CTO

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